LOCALISATIONS OF THE COMPLETION AT ASSOCIATED PRIMES OVER AN ANALYTICALLY UNRAMIFIED PRIME ARE DISCRETE VALUATION RINGS

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Allowed standard prerequisites (editorial).

- Analytic unramifiedness as reducedness of the completed quotient. Used at the first line of the counted proof (line 46120). Turns the hypothesis on p into reducedness of R-hat modulo p R-hat, which converts associated primes into minimal primes.
- Faithful flatness of the completion. Used at the end of the counted proof (lines 46128-46129). Lets nonzerodivisibility of an element be lifted from the completion back to R.

Lemma 0.1. *Let R be a Noetherian local ring. Let $\mathfrak{p} \subset R$ be a prime. Assume*

(1) *$R_\mathfrak{p}$ is a discrete valuation ring, and*
(2) *$\mathfrak{p}$ is analytically unramified.*

Then for any associated prime $\mathfrak{q}$ of $R^\wedge/\mathfrak{p}R^\wedge$ the local ring $(R^\wedge)_\mathfrak{q}$ is a discrete valuation ring.

Proof. Assumption (2) says that $R^\wedge/\mathfrak{p}R^\wedge$ is a reduced ring. Hence an associated prime $\mathfrak{q} \subset R^\wedge$ of $R^\wedge/\mathfrak{p}R^\wedge$ is the same thing as a minimal prime over $\mathfrak{p}R^\wedge$. In particular we see that the maximal ideal of $(R^\wedge)_\mathfrak{q}$ is $\mathfrak{p}(R^\wedge)_\mathfrak{q}$. Choose $x \in R$ such that $xR_\mathfrak{p} = \mathfrak{p}R_\mathfrak{p}$. By the above we see that $x \in (R^\wedge)_\mathfrak{q}$ generates the maximal ideal.

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As $R \to R^\wedge$ is faithfully flat we see that x is a nonzerodivisor in $(R^\wedge)_\mathfrak{q}$. Hence we win. $\square$